SMW-YOLO: Advancing Small Object Detection in Remote Sensing based on YOLOv11

Wenjing Lu*
School of Computer and Artificial Intelligence
Zhengzhou University
Zhengzhou, China
*quiet7525@163.com

Jiale Zhang
School of Computer and Artificial Intelligence
Zhengzhou University
Zhengzhou, China
zhangjl5594@gs.zzu.edu.cn

2025 6th International Conference on Computer Vision, Image and Deep Learning (CVIDL) | 979-8-3315-2324-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/CVIDL65390.2025.11085808

***Abstract*—This paper addresses the challenges of small target detection in remote sensing images, such as few target pixels, limited features, and complex backgrounds, by proposing an improved YOLOv11 model. The main improvements include: (1) designing the C3K2_StarsBlock module, which enhances the feature extraction of small targets through a star-shaped structure and feature aggregation; (2) proposing the MSAA module that integrates multi-scale features and attention mechanisms, combined with residual connections to improve model robustness; (3) using decoupled detection heads and the WIoU loss function to optimize classification and localization tasks. Experiments on the NWPU VHR-10 and DOTA datasets show that the proposed method improves mAP by 2.7% and 2.1% respectively, verifying its effectiveness.**

Keywords-Remote sensing images; small target detection; Multi-scale attention

I. INTRODUCTION

With the rapid development of optical remote sensing technology, small target detection has shown important application value in traffic monitoring,[1] search and rescue and other fields. However, small objects (usually less than 32×32 pixels) in remote sensing images have problems such as feature blur, low contrast and lack of information[7]. In addition, interference factors such as platform motion and atmospheric disturbance make the detection face three core challenges: insufficient feature representation ability, serious background interference and difficulty in multi-scale feature fusion. In response to the above challenges,[2][10]this study proposes an improved model based on YOLOv11. The main contributions include:

1) The C3K2_StarsBlock module is designed to enhance the feature extraction of small targets through star topology and multi-level feature aggregation.

2) The residual MSAA module is proposed, which integrates the multi-scale attention mechanism and the residual connection to improve the adaptability of the model to scale changes and complex backgrounds. 3) The decoupled detection head and WIoU loss function are used to optimize the performance balance between classification and localization tasks.

3) Decouple the detection head and WIoU loss function to optimize the performance balance of classification and positioning tasks.

This paper is structured as follows: Section II reviews related work; Section III details the model architecture. In Section 4, the effectiveness of the method is verified by comparative experiments. Section V concludes the paper and looks forward to future directions. Experiments show that the proposed method achieves 2.7%and 2.1%MAP improvement on NWPU VHR-10 and DOTA datasets, respectively, which is significantly better than the existing methods.

II. RELATED WORK

A. Application of YOLO in Remote Sensing Target Detection

The current object detection algorithms are mainly divided into two-level detectors (e.g., Faster R-CNN[13]) and one-level detectors (e.g., the YOLO series[2][3][4][5]). Compared with two-stage detectors, single-stage detectors like YOLO have faster inference speeds and are more suitable for real-time processing of remote sensing images. In recent years, researchers have proposed various improved YOLO algorithms for remote sensing images, such as FFCA-YOLO[8] with frequency-domain feature fusion,TPH-YOLO[11]with deformable convolution, and FE-YOLO[12] with deformable convolution. Although these methods improve detection performance, they often increase computational complexity. Therefore, based on the YOLOv11 framework, this paper designs a lightweight C3K2_StarsBlock and a residual MSAA module to enhance detection accuracy while ensuring real-time performance.

B. Feature Extraction and Enhancement of Small Targets

Aiming at the problem of limited feature information of small remote sensing targets, the existing research mainly improves in two directions: One is to enhance the feature expression ability through multi-layer feature fusion methods such as FPN[14] , PANet[15] and WBiFPN[9]; The second is to utilize the attention mechanism to enhance the weight of key features. The C3K2_StarsBlock module proposed in this paper innovatively combines the star topology structure and the multi-level feature aggregation mechanism, effectively enhancing the network's ability to capture the local details of small targets and significantly alleviating the problem of feature loss caused by the low pixel proportion.

C. Multi-scale feature fusion technology

The feature of large variations in the target scale in remote sensing images poses higher requirements for feature fusion. The traditional FPN[14] structure has limitations when dealing with complex backgrounds. Subsequent studies have made improvements by introducing additional supervision signals (AugFPN) or combining multi-depth pyramids (HRDNet), etc. The residual MSAA module proposed in this paper deeply integrates the multi-scale attention mechanism with the residual structure, and significantly improves the model's adaptability to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:15:08 UTC from IEEE Xplore. Restrictions apply.

scale changes through adaptive feature weighting. Meanwhile, the introduced dynamic upsampling mechanism further optimizes the feature fusion effect.

D. Loss Function Optimization

In view of the limitations of the traditional IoU loss in small object detection, this paper adopts the WIoU loss function. By dynamically adjusting the weights, WIoU can effectively reduce the influence of background interference and significantly improve the positioning accuracy of small targets. Compared with methods such as GIoU[17] and DIoU[16], WIoU has stronger robustness in complex backgrounds and is more suitable for remote sensing small target detection tasks

III. Method

This section proposes an improved SMW-YOLO network architecture based on YOLOv11, as shown in Figure 1. This architecture significantly improves the performance of small object detection through the following innovative designs:

Figure 1 shows our network architecture. The main improvements include: 1) Introduce the C3K2_StarsBlock module in the backbone network to enhance the feature extraction ability of small targets. 2) Design the residual MSAA module in the Neck part to improve the multi-scale feature fusion effect. 3) A decoupled detection head structure is adopted to alleviate the conflict between classification and positioning tasks. These improvements work together, significantly enhancing the performance of the model in the task of detecting small targets in remote sensing images. Furthermore, during the training process, we adopted the WIoU loss function and further optimized the positioning accuracy of the model in complex backgrounds by dynamically adjusting the weights, especially performing well in the task of small target detection.

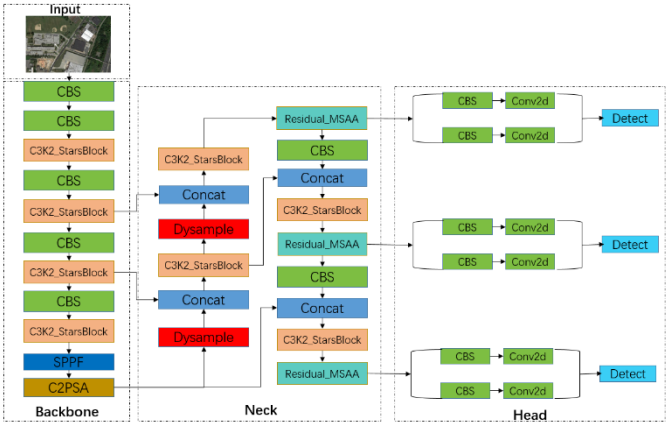


Figure 1. The network architecture of SMW-YOLO

A. The C3K2_StarsBlock module

Small targets in remote sensing images usually have the characteristics of low pixel proportion and limited feature information, which makes it difficult for traditional convolution operations to effectively extract the local detail information of the targets. To solve this problem, we designed the C3K2_StarsBlock module. This module combines the idea of STARNet and the backbone network structure of YOLOv11. By introducing the star topology structure and the multi-level feature aggregation mechanism, the feature extraction ability of the network for small targets is enhanced.

The C3K2_StarsBlock module is mainly composed of two key parts, and their structural diagrams are shown in Figures 2 and 3 respectively. Figure 2 shows the position of the module in the overall data processing flow and the initial data flow direction, while Figure 3 presents in detail the core composition structure inside the module.

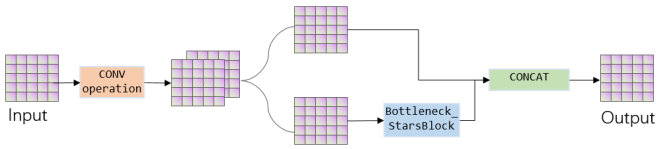


Figure 2. Overall architecture diagram of feature processing

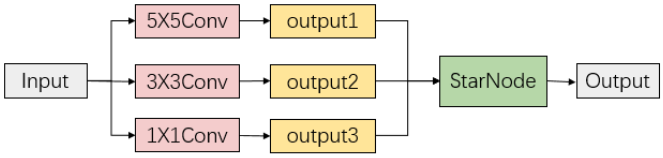


Figure 3. Multi-scale convolution branch and star node feature aggregation structure diagram

The C3K2_StarsBlock module adopts a multi-branch structure. Firstly, it undergoes standard convolution preprocessing, and then uses 1×1, 3×3, and 5×5 convolution in parallel to adjust the channels, extract local features, and capture context information respectively. By intelligently fusing multi-scale features through the star topology structure and finally splicing them with the features of another path, the ability to detect small targets can be improved while maintaining computational efficiency.

B. Residual MSAA module

The target scale in remote sensing images varies greatly and the background complexity is high, which puts forward higher requirements for the multi-scale feature fusion ability of the detection algorithm. To address this challenge, we designed the residual MSAA in the Neck part, deeply integrating the multi-scale attention mechanism with the residual structure. As shown in Figure 4, through adaptive weighted multi-scale features, the model's adaptability to scale differences was significantly improved, and the discrimination of target features in complex backgrounds was enhanced simultaneously.

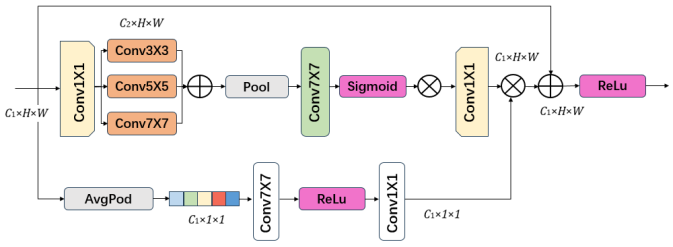


Figure 4. Residual MSAA structure diagram

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:15:08 UTC from IEEE Xplore. Restrictions apply.

The multi-scale feature fusion module proposed in this paper adopts a multi-branch parallel architecture. It extracts multi-scale features in parallel through 3×3, 5×5 and 7×7 convolution kernels, and innovatively introduces a dual attention mechanism (spatial attention highlights important regions, and channel attention optimizes feature weights). Finally, feature fusion is achieved through residual connection. This design effectively improves the detection performance of the network for targets of different scales, and significantly enhances the feature expression ability while maintaining good gradient propagation.

C. WIoU loss function

Let the prediction box be AA, the true box be BB, and the minimum circumscribed rectangle be CC. Then the traditional IoU calculation is Equation (1) :

$$\mathrm{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{1}$$

Define the background interference factor β :

$$\beta = \frac{|A \cap C| + |B \cap C|}{2|C|} \tag{2}$$

$|A \cap C|$ represents the overlapping area between the prediction box and the background.

Adopt the adaptive weights in exponential form:

$$w(\beta) = 1 + \lambda \cdot e^{\alpha\beta} \tag{3}$$

Among them, λ=0.5, λ=0.5 controls the weight amplitude, and α=2, α=2 regulates the sensitivity. The final loss function is:

$$\mathcal{L}_{WIoU} = w(\beta) \cdot (1 - \mathrm{IoU}) \tag{4}$$

This loss function has three significant advantages: 1) Scale awareness is achieved through the β factor, and small targets with a high background proportion are automatically identified; 2) The weight w(β) can be adaptively adjusted according to the background complexity; 3) Exponential design ensures the stability of the gradient magnitude. It can significantly improve the positioning accuracy of the model in the task of small object detection, especially performing well in scenarios with severe background interference.

IV. Experiment

A. Dataset

The experimental benchmarks used for model training and testing were selected from the NWPU VHR-10 and DOTA v1.5 datasets. The NWPU VHR-10 dataset contains 650 annotated images representing 10 types of ground objects, along with 150 background images. This dataset was randomly divided into independent training sets, validation sets and test sets in a ratio of 8:1:1. In contrast, the DOTA v1.5 dataset consists of 2,806 remote sensing images, covering 16 object categories of 188 scene types and nearly 400,000 objects, including small objects smaller than 10 pixels for measurement. This dataset is also divided into the training set, the validation set and the test set in a ratio of 6:2:2.

B. Hyperparameter setting

The experimental hyperparameters are set as shown in the following table:

TABLE I. Hyperparameter setting

Hyperparameter	Setting
Input resolution	640X640
Ir0	0.01
Lrf	0.01
momentum	0.937
weight_decay	0.0005
Batch_size	4
Epochs	200

C. Ablation experiment

To verify the effectiveness of each module proposed in this paper, we designed a step-by-step superimposed ablation experiment, and the results are shown in Table Ⅰ and Table Ⅱ. In the table, A stands for yolo11, C for C3K2_StarsBlock module, M for MSAA module, and W for WIOU.

TABLE II. Comparison of Ablation Experiment Results (NWPU VHR-10 Dataset)

Model	P	R	mAP0.5
A	89.3	88.7	90.2
A+C	90.5	88.8	90.5
A+M	91.4	88.5	90.5
A+W	90.8	88.9	90.9
A+C+M	91.7	89.2	91.1
A+M+W	91	88.6	91.6
A+C+M+W	92.6	90.6	92.9

TABLE III. Comparison of Ablation Experiment Results (DOTA Dataset)

Model	P	R	mAP0.5
A	78.7	78.9	78.3
A+C	79.8	79.8	79.2
A+M	80.2	79.9	79.1
A+W	79.9	80.0	79.5
A+C+M	79.4	79.7	78.9
A+M+W	80.1	80.2	79.3
A+C+M+W	81.8	80.9	80.4

Comprehensively considering Table Ⅱ and Table Ⅲ, it can be concluded that SMW-YOLO has achieved good results. Compared with the baseline model, SMW-YOLO provided a 2.7-percent-point increase in mAP@0.5 on the NWPU VHR-10 dataset and a 2.1-percent-point increase on the DOTA dataset. We also compared SMW-YOLO with other commonly used SOTA models, and the same training strategy was adopted in all experiments. The results of the comparative experiments are shown in the Table Ⅳ. The detection performance of the model SMW-YOLO on the NWPU VHR-10 dataset and DOTA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:15:08 UTC from IEEE Xplore. Restrictions apply.

dataset is higher than that of other models. The experimental parameter Settings are the same as those of the ablation experiment.[6] In the following table, D1 represents the NWPU VHR-10 dataset and D2 represents the DOTA dataset.

TABLE IV. COMPARISON EXPERIMENT RESULTS

Method	mAP@0.5(%)		mAP@0.5:0.95(%)	
	D1	D2	D1	D2
DETR	85.4	62.5	61.3	51.2
Deformable DETR	91.5	64.8	63.2	55.3
YOLOv5	83.8	71.6	57.2	57.6
YOLOv6	84.2	72.4	58.1	57.8
YOLOv8	89.8	78	59.8	57.3
YOLOv9	89.6	77.8	59.6	56.9
YOLOv10	89.7	77.6	59.7	57.1
YOLOv11	90.2	78.3	60.2	57.4
SMW-YOLO	92.9	80.4	61.2	58.8

V. CONCLUSIONS

In this paper, we proposed an improved YOLOv11 model to address the challenges of small target detection in remote sensing images, including limited pixel coverage, sparse features, and complex backgrounds. The key contributions include: (1) the C3K2_StarsBlock module, which enhances small target feature extraction through a star-shaped structure and feature aggregation; (2) the MSAA module, integrating multi-scale features and attention mechanisms with residual connections to improve robustness; and (3) the adoption of decoupled detection heads and the WIoU loss function to optimize classification and localization performance. Experimental results on the NWPU VHR-10 and DOTA datasets demonstrate the effectiveness of our method, achieving mAP improvements of 2.7% and 2.1%,respectively, compared to baseline models. The proposed approach effectively enhances small target detection accuracy while maintaining computational efficiency, making it suitable for practical remote sensing applications. Future work will focus on further optimizing real-time performance and generalizability across diverse scenarios.

REFERENCES

[1] M. Alqarqaz, M. Bani Younes, and R. Qaddoura, “An Object Classification Approach for Autonomous Vehicles Using Machine Learning Techniques,” World Electric Vehicle Journal, vol. 14, no. 2, p. 41, 2023.

[2] Varghese R, Sambath M. Yolov8: A novel object detection algorithm with enhanced performance and robustness[C]//2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS). IEEE, 2024: 1-6..

[3] Li, Chuyi,Li, Lulu,Jiang, Hongliang, et al.YOLOv6: A Single-Stage Object Detection Framework for Industrial Applications[J]. arXiv,2022,DOI:10.48550/arXiv.2209.02976.

[4] Wang C Y, Bochkovskiy A, Liao H Y M. YOLOv7: Trainable Bag-of Freebies Sets New State-of-the-Art for Real-Time Object Detectors[J]. 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022: 7464-7475.

[5] Wang C Y, Yeh I H, Liao H Y M. YOLOv9: Learning what you want to learn using programmable gradient information[C]//European Conference on Computer Vision. Cham: Springer, 2025.

[6] Zhu X , Su W , Lu L ,et al.Deformable DETR: Deformable Transformers for End-to-End Object Detection[C]//International Conference on Learning Representations.2021.

[7] V. Gagliardi et al., “Satellite remote sensing and non-destructive testing methods for transport infrastructure monitoring: Advances, challenges and perspectives,” Remote Sens., vol. 15, no. 2, p. 418, Jan. 2023.

[8] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo and J. Yan, "FFCA-YOLO for Small Object Detection in Remote Sensing Images," in IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1-15, 2024, Art no. 5611215, doi: 10.1109/TGRS.2024.3363057.

[9] L. Shen, B. Lang and Z. Song, "Object Detection for Remote Sensing Based on the Enhanced YOLOv8 With WBiFPN," in IEEE Access, vol. 12, pp. 158239-158257, 2024, doi: 10.1109/ACCESS.2024.3487492.

[10] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, ‘‘Object detection in optical remote sensing images: A survey and a new benchmark,’’ ISPRS J. Photogramm. Remote Sens., vol. 159, pp. 296–307, Jan. 2020, doi: 10.1016/j.isprsjprs.2019.11.023.

[11] X. Zhu, S. Lyu, X. Wang and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 Based on Transformer Prediction Head for Object Detection on Drone-captured Scenarios," 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), Montreal, BC, Canada, 2021, pp. 2778-2788, doi: 10.1109/ICCVW54120.2021.00312.

[12] Y. Zhang, H. Ma and Z. Xie, "Improved Fall-down Detection Algorithm: FE-YOLO," 2024 36th Chinese Control and Decision Conference (CCDC), Xi'an, China, 2024, pp. 2302-2306, doi: 10.1109/CCDC62350.2024.10587963.

[13] S. Ren, K. He, R. Girshick and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 39, no. 6, pp. 1137-1149, 1 June 2017, doi: 10.1109/TPAMI.2016.2577031.

[14] Lin, Tsung-Yi et al. “Feature Pyramid Networks for Object Detection.” 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2016): 936-944.

[15] Liu, Shu et al. “Path Aggregation Network for Instance Segmentation.” 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (2018): 8759-8768.

[16] Zheng, Zhaohui et al. “Distance-IoU Loss: Faster and Better Learning for Bounding Box Regression.” ArXiv abs/1911.08287 (2019): n. pag.

[17] Rezatofighi, Seyed Hamid et al. “Generalized Intersection Over Union: A Metric and a Loss for Bounding Box Regression.” 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2019): 658-666.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 06:15:08 UTC from IEEE Xplore. Restrictions apply.